

A Counterexample to AP Passing to Left-Ideal Quotients of Amenable Banach Algebras

FA Banach run fa_banach_001, agent agent_lane_10

2026-06-18

Status

This packet records a candidate counterexample, likely valid. It gives a negative answer to a formal question posed by Runde in arXiv:math/0203197: if A is an amenable Banach algebra with the approximation property, must A/L have the approximation property for every closed left ideal L ?

Source Passage

Runde asks the question after noting that the known Banach-space quotients of $\mathcal{L}^p$ -spaces without the approximation property do not carry any obvious Banach algebra structure.

- Both [14, I, Theorem 2.d.6] and [14, II, Theorem 1.g.4] are pure Banach space results, i.e. they involve no Banach *algebra* structure whatsoever. Hence, it may well be that case (ii) of Corollary 3.5 does never occur. In view of Proposition 1.1, we raise the following formal question: If $\mathfrak{A}$ is an amenable Banach algebra which has the approximation property, does $\mathfrak{A}/L$ have the approximation property for each closed left ideal L of $\mathfrak{A}$?

Transcription of the Question

The formal question on page 9 reads:

“If $\mathcal{A}$ is an amenable Banach algebra which has the approximation property, does $\mathcal{A}/L$ have the approximation property for each closed left ideal L of $\mathcal{A}$?”

Background Used

For a Banach space E , let $\mathcal{A}(E)$ denote the norm closure of the finite-rank operators on E . If E has the approximation property then $\mathcal{A}(E) = \mathcal{K}(E)$. Runde recalls that for $1 < p < \infty$, $p \neq 2$, the space ℓ_p has quotients which fail the approximation property [1]. We also use the standard fact, recorded for example in Blanco–Gronbaek and traced there to Johnson and to Gronbaek–Johnson–Willis, that $\mathcal{A}(\ell_p)$ is amenable for $1 < p < \infty$ [2].

Amenability is preserved by passing to the opposite Banach algebra: an approximate diagonal $\sum a_i \otimes b_i$ for B becomes $\sum b_i \otimes a_i$ for B^{op} .

The Counterexample

Theorem 1. *There is an amenable Banach algebra A whose underlying Banach space has the approximation property and a closed left ideal $L \subset A$ such that A/L does not have the approximation property.*

Proof Intuition

The obstruction in Runde's question is that pure Banach-space quotients ℓ_p/F without AP are not automatically quotients by algebraic left ideals. The construction puts them into that form by using an operator algebra. If $F \subset \ell_p$, the compact operators whose ranges lie in F form a right ideal in $\mathcal{A}(\ell_p)$. Passing to the opposite algebra turns this right ideal into a left ideal. The quotient by this ideal still remembers the Banach quotient ℓ_p/F : rank-one operators give a copy of ℓ_p/F , and evaluation at one vector gives a bounded projection back onto it. Thus a non-AP quotient of ℓ_p becomes a complemented subspace of a quotient of an amenable AP Banach algebra.

Proof. Choose $p \in (1, \infty) \setminus \{2\}$. By the Lindenstrauss–Tzafriri quotient results cited in Runde's remark, there is a closed subspace $F \subset \ell_p$ such that

$$Y = \ell_p/F$$

does not have the approximation property. Let $q : \ell_p \rightarrow Y$ be the quotient map.

Set

$$B = \mathcal{A}(\ell_p) = \mathcal{K}(\ell_p), \quad A = B^{\text{op}}.$$

The equality $\mathcal{A}(\ell_p) = \mathcal{K}(\ell_p)$ holds because ℓ_p has the approximation property. The algebra B is amenable for $1 < p < \infty$, and therefore so is A .

The Banach space $B = \mathcal{K}(\ell_p)$ has the metric approximation property. Indeed, let $P_n : \ell_p \rightarrow \ell_p$ be the coordinate projection onto the first n coordinates and define

$$\Phi_n(T) = P_n T P_n \quad (T \in \mathcal{K}(\ell_p)).$$

Each Φ_n has finite-dimensional range, $\|\Phi_n\| \leq 1$, and $\Phi_n(T) \rightarrow T$ in operator norm for every compact T . The last assertion follows because $P_n \rightarrow I$ uniformly on compact subsets of ℓ_p , and also $P_n^* \rightarrow I$ uniformly on compact subsets of $\ell_{p'}$, so

$$\|T - P_n T\| \rightarrow 0, \quad \|T - T P_n\| \rightarrow 0$$

for compact T .

Now define

$$J = \{T \in B : T(\ell_p) \subset F\}.$$

This is a closed right ideal of B : if $T(\ell_p) \subset F$ and $S \in B$, then $(TS)(\ell_p) \subset T(\ell_p) \subset F$. Hence J is a closed left ideal of the opposite algebra $A = B^{\text{op}}$. As Banach spaces,

$$A/J = B/J.$$

We show that B/J contains a complemented copy of Y . Choose $x_0 \in \ell_p$ and $f \in \ell_{p'} = \ell_p^*$ such that $f(x_0) = 1$. Define

$$i : Y \rightarrow B/J, \quad i(qx) = [x \otimes f],$$

where $(x \otimes f)(z) = f(z)x$, and brackets denote the coset modulo J . The map is well-defined: if $x - x' \in F$, then $(x - x') \otimes f$ has range contained in F , hence belongs to J . Moreover,

$$\|i(qx)\| \leq \|f\| \inf_{u \in F} \|x + u\| = \|f\| \|qx\|,$$

so i is bounded.

Define also

$$P : B/J \rightarrow Y, \quad P([T]) = q(Tx_0).$$

This is well-defined because $T \in J$ implies $Tx_0 \in F$. It is bounded, and for every $x \in \ell_p$,

$$Pi(qx) = P([x \otimes f]) = q(f(x_0)x) = qx.$$

Thus $Pi = \text{id}_Y$, so $i(Y)$ is a complemented subspace of B/J .

If $A/J = B/J$ had the approximation property, then its complemented subspace $i(Y)$ would have the approximation property. This is impossible because Y was chosen to fail AP. Therefore A/J fails the approximation property. The amenable Banach algebra A has AP, while its quotient by the closed left ideal J does not. This proves the claimed counterexample. $\square$

Verification Notes

The proof is analytic and uses no computer search. The included verification script checks the finite-dimensional matrix analogue: matrices with range contained in a subspace form a right ideal, and the rank-one/evaluation maps satisfy $Pi = \text{id}$ on the quotient coordinate space. This is only a sanity check for the algebraic orientation and complemented-copy mechanism.

Novelty and Literature Check

The run indexes were searched for arXiv:0203197, the source title, “reflexive amenable Banach algebra”, “amenable Banach algebra approximation property left ideal quotient”, “A(ell_p)”, and phrases around approximable operators and left-ideal quotients. The only direct run hit was an earlier no-promotion attempt for arXiv:0203197.

External web searches for exact phrases from Runde’s question and for later work on reflexive amenable Banach algebras did not reveal a direct settlement. A final targeted search for the exact counterexample shape, including “amenable Banach algebra closed left ideal fails the approximation property” and “A(ell_p) left ideal approximation property quotient”, returned no hits. The construction above uses a supporting source already in the local corpus, Blanco–Gronbaek arXiv:0804.1725, for the approximable-operator amenability background. Novelty confidence is moderate-to-high relative to the run memory: the example is short and uses standard ingredients, so human reviewers should still check whether this observation is known elsewhere.

Limitations

This is a counterexample to the formal AP-quotient question for closed left ideals. It does not settle Runde’s broader open problem of whether an infinite-dimensional amenable Banach algebra can have a reflexive underlying Banach space. The example $A = \mathcal{A}(\ell_p)^{\text{op}}$ is not reflexive.

Human Review Recommendation

Reviewers should check three points: the cited amenability of $\mathcal{A}(\ell_p)$, the MAP argument for $\mathcal{K}(\ell_p)$, and the orientation of the ideal after passing to the opposite algebra. The final obstruction is then immediate from AP passing to complemented subspaces.

References

- [1] V. Runde, *Banach space properties forcing a reflexive, amenable Banach algebra to be trivial*, arXiv:math/0203197, 2002.
- [2] A. Blanco and N. Gronbaek, *Amenability of algebras of approximable operators*, arXiv:0804.1725, 2008.
- [3] N. Gronbaek, B. E. Johnson and G. A. Willis, *Amenability of Banach algebras of compact operators*, Israel J. Math. **87** (1994), 289–324.
- [4] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces I and II*, Springer, 1977 and 1979.